

LIVERNE, MN, USA

WMO: 722006

Lat: 43.621N

Lon: 96.216W

Elev: 1431

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.3	-6.1	-17.8	2.2	-9.0	-13.4	2.8	-6.1	29.9	24.7	26.9	20.1	5.3	310	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	89.9	72.4	86.4	71.2	83.6	69.6	76.1	85.0	74.3	82.8	72.4	80.5	12.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	129.0	82.0	71.9	124.3	81.1	69.9	116.0	79.0	40.7	84.6	38.8	83.1	37.1	80.1	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.4	23.7	20.3	-20.9	94.1	5.9	2.6	-25.2	96.0	-28.7	97.5	-32.0	99.0	-36.3	100.8		
			-20.9	79.6	5.8	1.9	-25.1	80.9	-28.5	82.1	-31.8	83.1	-36.0	84.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	17.2	17.9	33.9	45.9	57.9	68.5	71.9	68.8	61.8	47.9	34.3	20.9
	DBStd	21.78	12.64	11.52	12.05	10.06	8.84	6.04	5.38	5.18	8.00	9.18	10.20	11.13
	HDD50	4183	1018	898	513	190	28	0	0	0	7	150	476	903
	HDD65	7631	1483	1318	964	574	254	32	9	21	153	535	920	1368
	CDD50	2624	0	0	14	68	273	556	679	583	361	85	5	0
	CDD65	597	0	0	0	2	34	137	223	138	58	5	0	0
	CDH74	5753	0	0	8	67	456	1325	2091	1147	585	74	0	0
	CDH80	1639	0	0	0	17	137	399	652	274	147	13	0	0
Wind	WSAvg	8.5	9.2	9.2	9.8	10.9	10.3	8.2	6.1	5.4	6.9	8.4	8.9	8.3
Precipitation	PrecAvg	27.2	0.6	0.7	1.4	2.8	3.5	4.4	3.4	3.2	2.9	2.2	1.2	0.8
	PrecMax	38.1	1.4	1.9	3.9	6.4	6.8	11.3	8.2	6.7	7.1	5.7	4.0	2.1
	PrecMin	17.1	0.1	0.2	0.1	0.5	1.0	1.3	0.4	1.3	0.3	0.4	0.0	0.1
	PrecStd	4.8	0.4	0.4	0.9	1.3	1.5	2.0	2.0	1.6	1.9	1.6	1.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.6	72.7	81.9	90.3	91.4	91.4	90.4	88.3	81.5	68.2	50.3
		MCWB	39.1	39.2	55.1	58.9	63.5	68.9	76.4	75.3	70.1	62.6	54.4	44.5
	2%	DB	37.5	40.8	64.0	73.2	82.4	88.2	90.0	86.2	84.2	73.5	61.0	42.9
		MCWB	34.3	36.0	52.7	55.2	63.1	69.1	75.3	73.9	68.7	58.4	49.5	37.3
	5%	DB	36.1	36.4	56.7	68.3	79.2	84.3	86.3	82.1	81.0	69.8	54.4	37.1
		MCWB	33.6	33.8	49.5	52.9	62.1	68.0	73.6	70.7	65.9	56.5	46.0	34.7
	10%	DB	33.9	34.0	50.5	63.2	73.3	81.4	82.4	80.9	77.1	63.5	49.8	35.7
		MCWB	32.5	32.2	45.4	50.2	58.5	66.2	70.6	69.7	64.4	52.9	43.3	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	40.5	59.4	61.5	70.5	75.4	80.0	77.9	74.0	67.2	56.6	45.0
		MCDB	45.0	45.5	68.2	73.1	80.5	86.1	88.8	86.2	82.9	74.9	64.9	49.7
	2%	WB	35.3	36.4	54.1	57.9	66.5	73.1	77.1	75.2	71.2	61.8	50.9	37.7
		MCDB	38.2	39.9	61.1	69.7	78.2	83.1	86.3	83.4	79.4	70.0	59.2	41.4
	5%	WB	33.5	33.9	50.0	55.2	63.9	71.0	75.1	73.2	69.1	57.6	46.5	35.3
		MCDB	35.0	36.1	58.1	64.8	74.2	80.5	83.8	80.6	77.8	67.0	53.4	37.6
	10%	WB	32.2	32.0	44.8	52.5	61.5	68.7	73.3	71.4	66.5	54.2	43.0	33.2
		MCDB	33.3	33.3	50.5	60.9	71.4	78.5	81.4	78.5	74.6	61.8	49.9	34.7
Mean Daily Temperature Range	5% DB	MDBR	16.9	17.2	17.3	20.6	21.5	20.3	20.8	20.3	23.4	21.4	19.3	15.8
		MCDBR	18.6	18.2	28.9	31.3	28.2	24.8	22.6	22.5	27.2	30.5	27.8	18.2
		MCWBR	15.8	14.9	17.9	16.1	12.9	11.1	11.8	12.4	13.0	16.2	17.4	14.5
	5% WB	MCDBR	16.6	17.3	27.4	26.1	23.1	20.4	20.6	20.2	22.4	24.8	25.9	17.2
MCWBR		14.5	14.6	17.3	15.3	12.0	11.3	11.7	12.2	12.3	14.6	17.0	13.8	
Clear-Sky Solar Irradiance	taub	0.267	0.282	0.315	0.370	0.390	0.400	0.408	0.411	0.361	0.325	0.290	0.274	
	taud	2.437	2.416	2.415	2.288	2.308	2.339	2.309	2.277	2.394	2.472	2.491	2.436	
	Ebn at Noon	279	292	294	284	280	277	272	268	274	272	267	264	
	Edh at Noon	23	28	32	40	40	39	40	40	33	27	22	21	
All-Sky Solar Radiation	RadAvg	601	940	1225	1534	1765	1977	2104	1739	1387	938	645	478	
	RadStd	56	76	98	139	128	128	79	130	100	114	43	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page